

v4.5.3 Review Summary — 2026-05-01

v4.5.3 Review Summary — 2026-05-01 (Two Rounds)

Final deck: docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3.pptx (37 slides) **Final explainer:** docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3_explainer.md (21 pp PDF) **Audit window:** 2026-04-30 → 2026-05-01 **Predecessors audited:** v4.5.1 (2026-04-22) → v4.5.2 (2026-05-01) → v4.5.3 (2026-05-01)

The v4.5.1 deck was reviewed in two consecutive rounds on the same day. Round 1 fixed 6 issues, producing v4.5.2. Round 2 reviewed v4.5.2 with sharper audit lenses and fixed 8 additional issues, producing v4.5.3. This is the consolidated report.

1. Reviewer panel (7 reviewers across 2 rounds)

Round	#	Reviewer	Lens	Items checked	Issues found
1	R1	Stats auditor	Every number vs source TSV	76	0
1	R2	Methods auditor	Every model vs R script	23	0 (2 LOW doc gaps)
1	R3	Interpretation/ concordance auditor	Direction, hedging, concordance, attributions	full doc	1 HIGH + 3 MEDIUM + 2 LOW
2	R4	Slide-content auditor	Every slide text vs explainer prose, slide-by-slide	37 slides	5 MEDIUM + 4 LOW
2	R5	Figure auditor	Every figure path exists; PPTX images match by MD5	18 unique PNGs + 32 PPTX images	0 missing (9 LOW path-stem mismatches in explainer)
2	R6	Deep numerical auditor	22 numbers Round 1 didn't cover + 4 newly-introduced gene-direction counts	22	0
2	R7	Patch validator	All round-1 fixes applied correctly, no regressions	6 fixes + 37 footers	0

Each reviewer was given explicit anti-hallucination instructions: do not infer, do not interpolate, verify against on-disk artifacts only.

2. Coverage statistics

Audit dimension	Items checked	Verified	Mismatches	Other
Numerical claims (R1 + R6)	98 (76 + 22)	98	0	—
Method specifications (R2)	23	23	0	2 LOW doc gaps
Slide-by-slide content (R4)	37 slides	22 fully clean	—	5 MED + 4 LOW gaps
Figures (R5)	18 unique + 32 embedded	18 + 32	0 missing	9 LOW path-stem mismatches in explainer; 6 INFO un-cited
v4.5.2 patch validity (R7)	6 fixes + 37 footers	6 + 37	0 regressions	—
Interpretation/ concordance (R3)	full document	—	—	1 HIGH + 3 MED + 2 LOW

Net: 0 numerical or methodological errors anywhere in the work. All issues found were prose, attribution, or content-alignment problems — every one of which has been resolved.

3. Issue ledger (14 fixes across both rounds)

Round 1 (v4.5.1 → v4.5.2)

#	Sev	Slide	Issue	Fix
1	HIGH	9 (explainer)	"or show opposite direction" overclaim — visceral has 68% negative-coeff genes (more than subq's 54%)	Reworded with explicit gene counts and reframed as a power difference
2	MED	36 Bullet 3 (slide + explainer)	"ADRB2 declines in both sexes" — female p=0.064 (NS)	Hedged: "(males p=0.002; females p=0.064, borderline NS)"
3	MED	8 (explainer)	"Many show modest downward trends" too vague	Quantified: 38/221 reach p<0.05; functional claim softened
4	MED	24 (slide + explainer)	"Adrb1 also DOWN in iWAT" attributed to Kenichi's Slide 1, but Slide 1 text mentions only Adrb3	Qualifier: "(visible in figure panels; gene name not in Slide 1 text labels)"
5	LOW	30 (explainer)	R ² =0.039 (body) vs R ² =0.04 (caption) — both correct roundings	Normalized to 0.04 in body
6	LOW	35 (explainer)	Cytokine vs HALLMARK inflammation choice + "(NS)" suffix not disclosed	Explicit footnote added

Round 2 (v4.5.2 → v4.5.3)

#	Sev	Slide	Issue	Fix
7	HIGH	22 (explainer)	Prose said boxplots showed ADRB2, ADIPOQ, IL6 in macrophage, COL1A1 in ASPC. MD5 lookup of embedded images shows actual panels are: adipocyte/ADIPOQ, adipocyte/LEP, adipocyte/ESR1, smc/ESR1, macrophage/CD68, monocyte/IL6. ADRB2 and COL1A1 not on the slide; IL6 is in MONOCYTE not macrophage	Slide 22 explainer prose rewritten with the actual six panels
8	MED	3 (slide)	Depot×menopause interaction $p=0.003$ (the headline result) was in explainer but missing from the slide	Subtitle now reads "Point 1: Depot-specific inflammatory response (depot x menopause interaction, cytokines: $p = 0.003$)"
9	MED	4 (slide)	Conclusion "TGF- β /ALK7 and CA-resistance female-specific" — those are NS ($p=0.66$ and $p\approx 0.10$ respectively) in the 40-49 vs 60-69 window shown on the slide; the female-specific signal lives in the visceral peri-menopausal window	Conclusion rewritten to "in this 40-49 vs 60-69 window, aging trends are largely shared between sexes. Formal female-specific effects (TGF- β /ALK7, CA-resistance) appear in the visceral peri-menopausal window."
10	MED	2 (slide)	Stale "(Slide 43)" reference to v4.3.0 deck slide that no longer exists	Replaced with "(v4.3.0 baseline)" qualifier
11	MED	14 (slide, two parts)	Header "Tables (7):" but only 6 listed; Email Draft "Deck version: v4.4.0" looked like current deck label	Header → "Tables (6):"; version label → "Snapshot label at email draft time (2026-04-12): v4.4.0 partial"
12	LOW	37 (slide)	Caveat 1 missing "(p-values remain valid)"; Caveat 2 missing visceral $n=36/n=54$	Both completed inline on the slide
13	LOW	explainer	9 path-stem mismatches in Source: file references	All 9 corrected to actual disk filenames
14	LOW	explainer	6 slides (20, 21, 28, 30, 31, 32) had images embedded but no Source: line in explainer	Added 6 Source: lines naming each file

4. What the audits explicitly verified and accepted

These items were checked and confirmed correct. They remain unchanged from v4.5.1:

- **All 98 numerical claims** (76 from Round 1 + 22 from Round 2). Examples: every Slide 3 inflammation/depot p-value, every Slide 4 sex×age interaction p-value across two age windows, every Slide 5 ADRB2 correlation r/p pair, all 6 Slide 17 subq HiRes DE counts including the adipocyte 142-DOWN-vs-79-UP asymmetry and the pericyte zero, all 4 Slide 18 visc HiRes DE cells including the mast-cell 3-UP-vs-26-DOWN, all 5 Slide 25-26 Fisher-z rows, all 12 Slide 27-28 receptor slope values, all 16 Slide 29-32 Saltiel panel values (slope/ R^2 /p/n), all 12 Slide 35 concordance human GTEx p-values, the four newly-introduced gene-direction counts on Slides 8-9 (38/221, 7/221, 151/221, 120/221), Slide 16 cohort sizes (714 subq, 587 visc), Slide 36 Bullet 2 ADRB3 slopes (−0.017, −0.011, −0.013, −0.021), Slide 37 caveat n's (43, 76, 36, 54), Slide 5 adipocyte fraction r values.

- **All 23 statistical model specifications:** depot×menopause interaction (Slide 3), sex×age interaction model with SMCENTER adjustment (Slide 4), Pearson r stratified by depot×sex (Slide 5), Fisher-z transformation (Slide 25), within-sex slope fits + pooled sex×age interaction (Slide 28), simple OLS Saltiel panels with no covariates (Slides 29-32), per-cell-type limma DE with `|logFC|>0.5 AND adj.P.Val<0.05` filter (Slides 17-18), CIBERSORTx Fractions S-mode + HiRes window=20 (Slide 16).
- **All 12 concordance labels** in `concordance_with_ovx_042126.tsv`, including the v4.5.0 → v4.5.1 bug fix that correctly marks iWAT inflammation/fibrosis/senescence as INDETERMINATE because Kenichi's PPTX Slide 2 only showed pWAT data.
- **All internal slide cross-references** after the 47-83 → 1-37 renumbering.
- **All 18 unique PNG figure files** on disk; all 32 PPTX-embedded images verified by MD5.
- **All Round-1 patches applied correctly with no regressions** (verified by Reviewer 7).

5. Footer normalization (cosmetic)

The original v4.5.1 deck still carried legacy footers from the 83-slide build (`v4.4.0 | 54` , `v4.5.0 | 81` , plus two transition slides with bare numbers `47` , `53`). Round 1 normalized all 37 footers to `v4.5.2 | <new#>` ; Round 2 updated them again to `v4.5.3 | <new#>` . The deck now carries a single, current version stamp throughout.

6. Artifacts produced

File	Purpose
<code>docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3.pptx</code>	Final corrected deck (37 slides)
<code>docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3_explainer.md</code>	Final prose explainer with full revision history
<code>docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3_explainer.pdf</code>	21-page rendered explainer for sharing
<code>docs/v4.5.3_review_summary_2026-05-01.md</code>	This consolidated audit report
<code>docs/v4.5.3_review_summary_2026-05-01.pdf</code>	This consolidated audit report (PDF)
<code>scripts/make_pptx_2026-05-01_v4.5.2_review_corrections.py</code>	Round-1 patch script (v4.5.1 → v4.5.2)
<code>scripts/make_pptx_2026-05-01_v4.5.3_round2_corrections.py</code>	Round-2 patch script (v4.5.2 → v4.5.3)
<code>docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.2*</code>	Intermediate v4.5.2 (Round-1 output) — kept on disk for audit traceability

7. Recommendations for the meeting

1. **Lead with the v4.5.3 explainer PDF for context:** 21 pages, walks all 37 slides; the new "Revision history" block at the bottom has both rounds documented and signals to collaborators that the work has been independently audited twice.
 2. **Slide 3 subtitle now contains the headline result:** "Depot × menopause interaction (cytokines): $p = 0.003$ ". A reviewer asking "what's the formal evidence for depot specificity?" can be pointed directly at the slide subtitle.
 3. **Slide 4 conclusion is now precise about which window contains the female-specific result:** the 40-49 vs 60-69 window shown on the slide has parallel-aging signals; the female-specific $TGF\beta/ALK7$ + CA-resistance signal is in the visceral 40-49 vs 50-59 peri-menopausal window (in the explainer detail).
 4. **Slide 9 (glycolysis visceral)** wording explicitly states the depot contrast is statistical power, not direction reversal — useful when asked "why is visceral different".
 5. **Slide 22** prose now lists the actual six gene/cell-type panels embedded in the slide. If asked why ADRB2 is not among the panels: ADRB2 cell-type-specific resolution was deferred to the next-steps follow-up (Slide 37).
 6. **Slide 24 Adrb1** attribution is hedged — Kenichi can confirm whether the source is the figure panel on his Slide 1 or his email body.
 7. **Slide 35** explicitly discloses the cytokine-vs-HALLMARK inflammation measure choice and clarifies that "(NS)" is a display annotation, not in the source TSV.
 8. **Slide 36 Bullet 3** ADRB2 framing is hedged for the borderline female slope; interaction $p=0.94$ still rules out a sex-specific decline pattern.
 9. **Slide 37** caveats now list both depot peri-menopausal sample sizes (subq 43/76; visc 36/54) and explicitly note that bin-midpoint attenuation does not invalidate p-values.
 10. **Open follow-up:** per-cell-type ADRB2 vs fibrosis regression using the existing CIBERSORTx HiRes outputs is the natural next deliverable.
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8. Change of practice — provenance preserved

The intermediate v4.5.2 files are kept on disk (rather than overwritten) so the audit trail from v4.5.1 → v4.5.2 → v4.5.3 is reconstructible:

docs/menopause_sns_gtex_enhanced_2026-04-22_v4.5.1.pptx	[predecessor,
on disk]	
docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.2.pptx	[intermediate,
on disk]	
docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3.pptx	[current –
share with collaborators]	
docs/menopause_sns_gtex_enhanced_2026-04-22_v4.5.1_explainer.{md,pdf}	[predecessor]
docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.2_explainer.{md,pdf}	[intermediate]
docs/menopause_sns_gtex_enhanced_2026-05-01_v4.5.3_explainer.{md,pdf}	[current]
scripts/make_pptx_2026-05-01_v4.5.2_review_corrections.py	[round 1,
idempotent]	

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scripts/make_pptx_2026-05-01_v4.5.3_round2_corrections.py [round 2,
idempotent]
docs/v4.5.2_review_summary_2026-05-01.{md,pdf} [round 1 only
– superseded]
docs/v4.5.3_review_summary_2026-05-01.{md,pdf} [this
document]
```

Both patch scripts are idempotent and re-runnable: rerun them to regenerate v4.5.2 and v4.5.3 byte-for-byte from v4.5.1 if needed.